

Overview

In this topic, you will learn about a greenhouse gas (GHG) account and how it quantifies and records your GHG emissions and carbon storage.

The topic starts by explaining GHG accounting and the reasons for doing it. It examines the benefits of a farmer or land manager having a GHG account. It also examines accounting for co-benefits and the uses of a GHG account.

Keeping a GHG account is about estimating GHG emissions and carbon storage using a GHG calculator and recording these estimates. There are many calculators available. Some make quick, simple estimates that give a general understanding and allow users to run ‘what if?’ scenarios. Others are more complex and make more precise estimates where needed, for example, to earn ACCUs. This topic explains 3 calculators, covering various sectors and levels of complexity and gives farmers and land managers hands-on experience using them. The 3 calculators are:

- the Meat and Livestock Australia (MLA) Quick Start Carbon Calculator, a simple calculator that estimates GHG emissions and carbon storage that only needs basic data
- the Greenhouse Accounting Framework (GAF) tools, a group of agriculture-specific tools produced by the Primary Industries Climate Challenges Centre and the University of Melbourne
- the Commonwealth Scientific and Industrial Research Organisation (CSIRO) LOOC-C (Landscape Options and Opportunities for Carbon abatement Calculator), which gives estimates and options for action linked to the ACCU Scheme.

Section 7 provides information on other calculators, including a table summarising them, to help in considering their suitability for a particular purpose.

Farmers and land managers often already have the data they need to use these calculators or are likely to be able to collect the data. For example, a farmer might have the data for farm input benchmarking or management or tax accounting.

Learning outcomes

After completing this topic, you will be able to:

- describe GHG accounts and the need for them
- identify 3 calculators and what data they need
- calculate GHG emissions and carbon storage for a property using a GHG calculator
- identify the largest sources of emissions and potential options to reduce emissions or store carbon.

Watch this video

In this video (2:10 minutes), presenters Gail Reynolds-Adamson and Matt Woods introduce Topic 3 and provide important context.

GAIL REYNOLDS-ADAMSON: Welcome to topic three, where we look at understanding emissions in more detail, and in particular, how to accurately measure them.

MATT WOODS: Einstein supposedly once said, "Not everything that can be counted counts, and not everything that counts can be counted". Now he obviously wasn't talking about carbon farming because measuring emissions are what counts, and they can be counted.

The key is knowing how much emissions are going out and how much are being avoided or saved.

We call this carbon accounting.

REYNOLDS-ADAMSON: And carbon accounting is important when carbon farming. Whether you intend to sell carbon credits or not, being able to establish a baseline measure and report your emissions is vital for the Australian Carbon Credit Unit, ACCU, schemes.

But equally, for demonstrating to supply chains and to customers the fruits of your carbon farming endeavour. The key is knowing what to measure and how, and this is where this topic will help. Like most things agricultural, it's not always straightforward, and there's many different tools and techniques with various pros and cons.

Sometimes it can be hard to sort the wheat from the chaff.

WOODS: This topic will cover the on farm information needed to measure emissions. We'll look at various carbon calculators, how to choose the right ones for your sector, and the kinds of data they need.

REYNOLDS-ADAMSON: Again, you will hear from experts and see relevant case studies and be prompted to respond to the focus questions in preparation for discussions with the experts to guide your decisions.

And one more thing before you go.

Whilst we can and must measure emissions and other co-benefits such as biodiversity, there are some that, whilst they count, simply cannot be counted.

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/climatechange/carbon-farming-outreach-program/training-package/topic-3/1-overview-learning-outcomes>